EE526/RA506 Machine Learning

End-Semester Exam

Date: 03 May 2026 Full Marks: 150 Time: 3 Hours

1. A sensor is receiving samples of a gradually decaying signal of the form $x[n] = A\beta^n$ ($n \geq 0$, $0 < \beta < 1$). The sensor is feeding the data to a system that incrementally updates the mean as $\mu[n] = (1-\alpha)\mu[n-1] + \alpha x[n]$. Here, $\alpha \in (0,1)$ is a fixed update rate and $\alpha \neq \beta$. Evaluate $\mu[n]$ for this input $x[n]$ where $\mu[0] = A$. [15]

2. Let C_n be the instantaneous estimate of the Covariance Matrix of streaming vector data. Deduce the expression for updating C_{n-1} to C_n by using a fixed update rate α $(0 < \alpha < 1)$ and $\mathbf{x}_n$ (vector data at n^{th} instant). [10]

3. Consider a SLFN with input layer $\mathbf{X} \in \mathbb{R}^2$, hidden layer $\mathbf{H} \in \mathbb{R}^3$ (*Swish* activation), and output layer $\mathbf{Y} \in (0,1)^2$ (*SoftMax* activation).

$$W_1 = \begin{bmatrix} -0.14 & 0.21 \\ 0.09 & 0.19 \\ 0.55 & -0.75 \end{bmatrix}, \quad b_1 = \begin{bmatrix} 0.48 \\ 0.07 \\ -0.07 \end{bmatrix}, \quad W_2 = \begin{bmatrix} -0.17 & 0.11 & 0.11 \\ 0.88 & -0.17 & 0.53 \end{bmatrix}, \quad b_2 = \begin{bmatrix} 0.18 \\ 0.33 \end{bmatrix}$$

Evaluate $\mathbf{H}$ and $\mathbf{Y}$ for input $\mathbf{X} = [-7 \quad 8]^T$ and the SLFN parameters (W_1, b_1) between $\mathbf{X} \to \mathbf{H}$, and (W_2, b_2) between $\mathbf{H} \to \mathbf{Y}$. [15]

4. Consider an MLP with input $\mathbf{X} \in \mathbb{R}^D$, hidden layers $\mathbf{H}_1 \in \mathbb{R}^K$ (*Swish* activation), $\mathbf{H}_2 \in (-1,1)^L$ (*tanh* activation), and output $\mathbf{Y} \in (0,1)^M$ (*SoftMax* activation). The parameters between $\mathbf{X} \to \mathbf{H}_1$, $\mathbf{H}_1 \to \mathbf{H}_2$, and $\mathbf{H}_2 \to \mathbf{Y}$ are (W_1, B_1), (W_2, B_2), and (W_3, B_3) respectively. Report the (a) total number of trainable MLP *Parameters* (Params), (b) *Feedforward Multiplications* (MUL), (c) *Feedforward Additions* (ADD), (d) *Feedforward Divisions* (DIV), (e) *Feedforward Exponential Evaluations* (EXP), and (f) *Feedforward Comparisons* (CMP). Also, deduce the expression for $\frac{\partial \mathcal{L}_{CE}}{\partial W_1(i,j)}$ for weight update by gradient descent while minimizing the *Cross-Entropy Loss* $\mathcal{L}_{CE}$ over the training dataset $\mathbf{D}_{trn} = \{(\mathbf{X}_n, \mathbf{Y}_n); n = 1, \ldots N\}$. **[2.5 × 6 + 15 = 30]**

5. Perform kernel-based regression for the multivariate function $\mathbf{y} = \mathbf{F}(\mathbf{x})$, where $\mathbf{x}, \mathbf{y} \in \mathbb{R}^2$. The following table lists 10 data instances $(\mathbf{x}, \mathbf{y})$. Using these instances, report the estimate of $\mathbf{y}$ at $\mathbf{x} = [1.38 \;\; 1.50]^T$. Use an Epanechnikov kernel with bandwidth $h = 0.19$. [20]

	Sample index									
	1	2	3	4	5	6	7	8	9	10
x_1	1.35	1.17	1.30	1.59	1.52	1.27	1.41	1.28	1.61	1.35
x_2	1.60	1.38	1.55	1.47	1.37	1.56	1.73	1.42	1.44	1.39
y_1	0.44	0.09	0.02	−0.15	0.54	−0.35	−1.11	−0.74	−0.34	0.27
y_2	0.21	0.01	0.00	−0.05	−0.05	0.14	1.41	0.27	−0.51	−0.02

6. The instances in the multi-category dataset $\mathbf{S_m}$ consisting of 5 different classes are present in the proportion of 23 : 7 : 11 : 17 : 13. Report the *Misclassification* (H_m), *Entropy* (H_e) and *Gini* (H_g) impurities of dataset $\mathbf{S_m}$. [10]

7. The dataset $S = S^+ \bigcup S^-$ is presented to a node of a classification tree. Here, *Positive Class* $S^+ = \{(10, 8), (-7, -2), (-2, 6), (-3, 9), (2, 2)\}$ and *Negative Class* $S^- = \{(10, -5), (12, 3), (1, -6), (8, 2), (9, -5)\}$. Split this dataset S by maximizing the misclassification impurity drop. Report (a) the optimal decision rule in the form $X[j^\star] \geq \theta^\star$ (threshold θ is taken from data values); and (b) resultant impurity drop ΔH. [30]

8. Group the dataset $\mathbf{D}$ into $K = 3$ clusters by K-means clustering. Iterate till convergence and report the data labels (L_t) and cluster centroids (μ_1, μ_2, μ_3) for each iteration t in the following table. [20]

$\mathbf{D} \rightarrow$	5	33	37	8	65	9	67	40	6	18	μ_1	μ_2	μ_3
L_0	1	2	3	1	2	3	1	2	3	1			
L_1													
L_2													
L_3													
L_4													
L_5													
L_6													